

Project Brand: TERP PROTECT

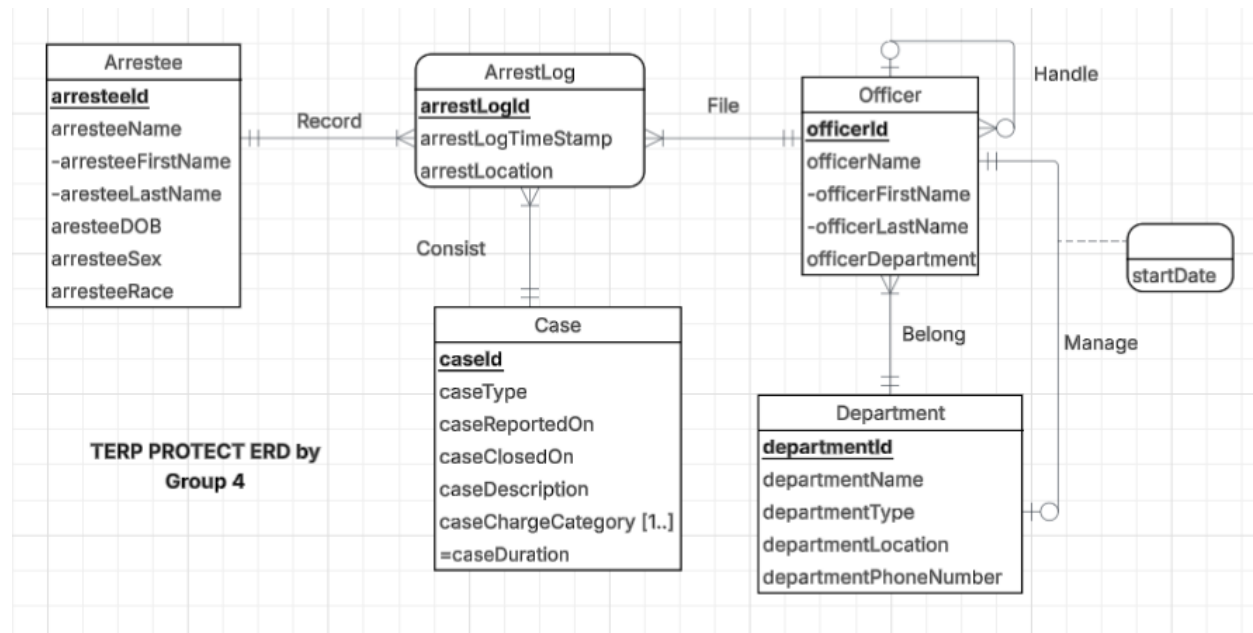
Mission Statement

To design a database system that organizes and manages Arrest logs, allowing the University of Maryland Police Department to analyze arrests by demographics, Charges, Officers, and Locations.

Mission Objectives

- 1. Which age and sex groups have the highest number of arrests?**
 - 2. Determine the most common charge categories leading to arrests and how they vary over time.**
 - 3. Calculate the number and proportion of reported cases that did not result in any arrest.**
 - 4. Compute and compare the average case duration across departments to identify performance differences.**
 - 5. Identify which departments and department heads have recorded the highest number of arrests and handled the most cases.**
 - 6. Determine the most frequently handled case types by each department.**
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ER Model



Entities for Arrest Log Database

1. Case

- **Why:** Every arrest belongs to a case; some cases may have no arrests.
- **Attributes:**
 - Case ID (PK)
 - Case Type
 - DateReported On
 - Description
 - Case Charge Category
 - Case Closed On
 - Case Duration(Derived Attribute)

2. Arrest Log

- **Why:** Central event tying case, suspect, officer, and charges.
- **Attributes:**
 - Arrest ID (PK)

- Arrest Log Time Stamp
- Arrest Location
- Case ID (FK → Case)
- Arrestee ID (FK → Arrestee)
- Officer ID (FK → Officer)

3. Arrestee

- **Why:** For demographic analysis (age, sex, race, repeat offenders).
- **Attributes:**
 - Arrestee ID (PK)
 - Arrestee Name
 - Arrestee DOB (or Age)
 - Arrestee Sex
 - Arrestee Race

4. Officer

- **Why:** Needed for accountability and performance tracking.
- **Attributes:**
 - Officer ID (PK)
 - Officer Name
 - Officer Department

5. Department

- **Why:** To organize officers by their assigned departments and track departmental information.
 - **Attributes:**
 - departmentId (PK)
 - departmentName
 - departmentType
 - departmentLocation
 - departmentPhoneNumber
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Relationships

1. Arrestee – ArrestLog

- One Arrestee can have many ArrestLogs (0..*). Each ArrestLog belongs to exactly one Arrestee (1).
- **Cardinality:** (Arrestee) 1 : * (ArrestLog)

2. Case – ArrestLog

- *One or many cases can have one or many ArrestLogs. Each ArrestLog must belong to exactly one Case.*
- **Cardinality:** (Case) 1 : * (ArrestLog)

3. Officer – ArrestLog

- *One Officer can file many ArrestLogs. Each ArrestLog is recorded by exactly one Officer.*
- **Cardinality:** (Officer) 1: * (ArrestLog)

4. Officer → Department (Belongs)

- *A department can contain 1 or more officers. Each officer belongs to a single department.*
- **Cardinality:** (Officer) * : 1 (Department)

5. Officer → Department (Manages)

- *A department is headed by a single manager*
- **Cardinality:** (Officer) 1: 1 (Department)

6. Officer → Officer(Handle)

- *A department is headed by a single manager*
- **Cardinality:** (Officer) 1: 1 (Department)

Relational Schema

Arrestee(arresteeld, arresteeFirstName, arresteeLastName, arresteeDOB, arresteeSex, arresteeRace)

Officer(officerId, officerFirstName, officerLastName, *departmentId*, *officerLeadId*)

Department(departmentId, departmentName, departmentType, departmentLocation, departmentPhoneNumber, *officerHeadId*, startDate)

Case(caseId, caseType, caseReportedOn, caseClosedOn, caseDescription)

caseChargeCategory(*caseId*, caseChargeCategory)

ArrestLog(*caseId*, *arresteeld*, *officerId*, arrestLogId, arrestLogTimeStamp, arrestLocation)

Business Rules

[R1] When a department is deleted from the database, the deletion will be prevented if there are any officers assigned to that department.

[R2] When a department is updated, all officers assigned to that department should automatically reflect the updated department information accordingly.

[R3] When a lead officer is deleted from the database, the deletion will be prevented if there are any officers reporting to that lead officer.

[R4] When a lead officer's information is updated, the update will be prevented if there are any officers reporting to that lead officer.

[R5] When an officer who serves as a department head is deleted from the database, the deletion will be prevented if that officer is currently assigned as head of any department

[R6] When a department head officer's information is updated, the department record should automatically reflect the updated officer information accordingly.

[R7] When a case is deleted from the database, all case charge categories associated with that case should also be deleted. When case information is updated, all related case charge categories should reflect those changes accordingly.

[R8] When a case is deleted from the database, all arrest logs linked to that case should also be deleted. When case information is updated, all arrest logs associated with that case should reflect those changes accordingly.

[R9] When an arrestee is deleted from the database, all arrest logs for that arrestee should also be deleted. When arrestee information is updated, all arrest logs should reflect the updated arrestee information accordingly.

[R10] When an officer is deleted from the database, all arrest logs filed by that officer should also be deleted. When officer information is updated, all arrest logs filed by that officer should reflect the updated officer information accordingly.

Referential Integrity Actions:

Relation	Foreign Key	Base Relation	Primary Key	Business Rule	ON DELETE	Business Rule	ON UPDATE
Officer	departmentId	Department	departmentId	R1	NO ACTION	R2	CASCADE
Officer	officerLeadId	Officer	officerId	R3	NO ACTION	R4	NO ACTION
Department	officerHeadId	Officer	officerId	R5	NO ACTION	R6	CASCADE
caseChargeCategory	caseId	Case	caseId	R7	CASCADE	R7	CASCADE
ArrestLog	caseId	Case	caseId	R8	CASCADE	R8	CASCADE
ArrestLog	arresteeld	Arestee	arresteeld	R9	CASCADE	R9	CASCADE
ArrestLog	officerId	Officer	officerId	R10	CASCADE	R10	CASCADE

Sample Data

Arrestee Table

arresteeId	arresteeFirstName	arresteeLastName	arresteeDOB	arresteeSex	arresteeRace
A101	John	Miller	12/04/90	M	White
A102	Priya	Singh	22/09/87	F	Asian
A103	Marcus	Brown	18/01/95	M	Black
A104	Sofia	Gonzalez	30/03/98	F	Hispanic
A105	Ethan	Clark	14/07/00	M	White
A106	Olivia	Turner	11/08/93	F	Black
A107	Daniel	Kim	22/06/92	M	Asian
A108	Emma	Lewis	09/10/97	F	White
A109	Carlos	Rivera	15/12/89	M	Hispanic
A110	Maya	Patel	17/02/96	F	Asian

Officer Table

officerId	officerFirstName	officerLastName	departmentId
O201	David	Carter	D101
O202	Linda	Johnson	D102
O203	Kevin	Patel	D101
O204	Maria	Lopez	D103
O205	Robert	Wilson	D102
O206	Sarah	Thompson	D104
O207	James	Nguyen	D105
O208	Lisa	Park	D104

O209	Michael	Adams	D105
O210	Hannah	Wright	D103
officerId	officerFirstName	officerLastName	departmentId
O201	David	Carter	D101
O202	Linda	Johnson	D102
O203	Kevin	Patel	D101
O204	Maria	Lopez	D103

Case Table

caseId	caseType	caseReportedOn	caseClosedOn	caseDescription
C301	Cyber Fraud	11/01/23	20/02/23	Online scam involving stolen credit cards
C302	Drug Possession	02/05/23	15/06/23	Suspect caught with illegal narcotics
C303	Identity Theft	10/07/23	25/09/23	Unauthorized use of personal data
C304	Homicide	30/11/22	18/04/23	Murder investigation in downtown area
C305	Drug Trafficking	22/03/23	05/08/23	Cross-state drug transport case
C306	Hit and Run	12/02/23	22/02/23	Driver fled the scene after collision
C307	Financial Scam	01/09/23	30/09/23	Ponzi scheme under investigation
C308	Kidnapping	25/05/23	01/08/23	Missing person traced through phone records
C309	Cyber Hacking	14/06/23	20/07/23	Corporate server breach incident
C310	Tax Evasion	05/12/22	10/05/23	Large-scale financial fraud detected

Case Charge Category Table

caseId	caseChargeCategory
C301	Felony
C302	Misdemeanor
C303	Felony
C304	Felony
C305	Felony
C306	Misdemeanor
C307	Felony
C308	Felony
C309	Felony
C310	Felony

Department Table

departmentId	department Name	department Type	departmentLocation	departmentPhone Number	officerHeadId	startDate
D101	Cyber Crimes	Investigation	New York HQ	212-555-8711	O201	12/05/18
D102	Narcotics	Enforcement	Los Angeles	310-555-9923	O202	18/03/16
D103	Homicide	Investigation	Chicago	773-555-1176	O204	21/09/15
D104	Traffic Control	Field Operations	Miami	305-555-6632	O206	05/01/19
D105	Financial Crimes	Investigation	Houston	832-555-8819	O207	22/07/17
D106	Forensics	Support	New York HQ	212-555-4409	O201	10/02/20

D107	Cyber Security	Prevention	San Francisco	415-555-2277	O203	13/11/19
D108	Missing Persons	Investigation	Dallas	469-555-6690	O204	09/12/16
D109	Fraud Detection	Investigation	Seattle	206-555-7788	O207	19/08/18
D110	Emergency Response	Field Ops	Denver	303-555-9876	O206	25/04/21

Arrest Log Table

arrestLogId	caseId	arrestedId	officerId	arrestLogTimeStamp	arrestLocation
AL401	C301	A101	O203	11/01/23	Manhattan
AL402	C302	A103	O205	02/05/23	Los Angeles Downtown
AL403	C303	A102	O203	10/07/23	Brooklyn
AL404	C304	A104	O204	30/11/22	Chicago Loop
AL405	C305	A105	O205	22/03/23	Santa Monica
AL406	C306	A106	O208	12/02/23	Miami Beach
AL407	C307	A107	O209	01/09/23	Houston Heights
AL408	C308	A108	O210	25/05/23	Chicago Southside
AL409	C309	A109	O203	14/06/23	San Francisco HQ
AL410	C310	A110	O209	05/12/22	Houston Uptown

The sample data represents a law enforcement database where each table works together to track arrests and investigations. The Arrestee relation stores personal details about individuals who have been arrested, including their name, date of birth, sex, and race. The Officer relation contains details about police officers, their assigned department, and their supervising officer. Each Department record identifies a specialized unit (like Cyber Crimes or Narcotics), its location, contact information, and the head officer managing it. The Case relation tracks incidents under investigation, recording the type of case, when it was reported and closed, and a short description. The CaseChargeCategory specifies the legal classification (such as felony or misdemeanor) linked to each case. Finally, the ArrestLog acts as a junction table connecting arrestees, officers, and cases, recording when and where each arrest occurred. Together, this data flow shows how departments and officers investigate cases that lead to arrests, with every event traceable through interconnected entities.